

Q1:

a.)

for a system with $L_1 \dots L_n, S_1 \dots S_n$

it first couple the L and S

that

$$\vec{L} = \sum_i \vec{L}_i \quad \vec{S} = \sum_i \vec{S}_i$$

than couple total L & S into J

$$\vec{J} = \vec{L} + \vec{S}$$

LS applies spin orbit coupling between L & S
are weak

$$3s^2 \ ^1S \quad 3s^2 \ ^3S$$

$$3s3p \ ^1P \quad 3s3p \ ^3P$$

$$3s3d \ ^1D \quad 3s3d \ ^3D$$

$$b-) \vec{\mu} = -\frac{\mu_B}{\hbar} (\vec{L} + 2\vec{S})$$

$$E = -\langle \vec{\mu} \cdot \vec{B} \rangle = -\langle \vec{\mu}_{\text{eff}} \cdot \vec{B} \rangle$$

$$\text{in LS, } L, S \text{ couple into } J \text{ so } \vec{\mu}_{\text{eff}} = \frac{(\vec{\mu} \cdot \vec{J}) \vec{J}}{J^2}$$

which is projection of μ_{eff} in $\vec{J}$ direction

$$\vec{\mu} = \frac{\mu_B}{\hbar} (\vec{L} + 2\vec{S}) \cdot \vec{B}$$

$$\Delta E = g_J \mu_B B_z M_J$$

$$\vec{L} + 2\vec{S} = \frac{(\vec{L} \cdot \vec{J})\vec{J}}{J^2} + 2 \frac{(\vec{S} \cdot \vec{J})\vec{J}}{J^2}$$

~~$$\vec{L} \cdot \vec{J} = S = J - L, \quad S^2 = J^2 + L^2 - 2\vec{L} \cdot \vec{J}$$~~

$$\vec{L} \cdot \vec{J} = \frac{1}{2}(J^2 + L^2 - S^2)$$

$$L = J - S \quad L^2 = J^2 + S^2 - 2\vec{J} \cdot \vec{S}$$

$$\vec{S} \cdot \vec{J} = \frac{1}{2}(J^2 + S^2 - L^2)$$

$$H = \frac{\mu_B}{\hbar} \left[\frac{J^2 + L^2 - S^2 + 2J^2 + S^2 - 2L^2}{2J^2} \right] \vec{J} \cdot \vec{B}$$

$$g_J = \frac{3J^2 + S^2 - L^2}{2J^2}$$

$$\Delta E = \frac{\mu_B}{\hbar} M_J B_z \left\langle \frac{3J^2 + S^2 - L^2}{2J^2} \right\rangle$$

$$= \frac{\mu_B}{\hbar} M_J B_z \frac{3J(J+1) + S(S+1) - L(L+1)}{2(J+1)J}$$

$$g_J$$

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}$$

c.)

$$\Delta E = \frac{\mu_B B}{\hbar} g_J = g_J \mu_B B$$

for A: $\Delta E = h \nu$

$$= 6.96 \times 10^{-24} \text{ J}$$

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$$g_J = \frac{\Delta E \cdot \hbar}{\mu_B B} \quad g_J = \frac{\Delta E}{\mu_B B} \sim 0.5$$

3 level $\rightarrow M_J = -1, 0, 1$, $J=1$

for C: 9 level $(2J+1)=9 \Rightarrow J=4$

$$\Delta E = 6.96 \times 10^{-24} \quad g_J \sim 0.5$$

for A:

$$\frac{1}{2} = \frac{S(S+1) - L(L+1) + 6}{4}$$

$$S(S+1) - L(L+1) = -4$$

$$\Rightarrow \underline{S=1 \quad L=2}$$

for C:

$$\frac{1}{2} = \frac{S(S+1) - L(L+1) + 60}{40}$$

$$S(S+1) - L(L+1) = -40$$

$$\Rightarrow \underline{S=1, L=6}$$

$$d.) \quad 2J+1=6 \quad J=5/2, \quad g_J=1/2$$

$$\frac{1}{2} = \frac{5(5+1) - 1(1+1) + 3 \cdot \frac{5}{2} \cdot \frac{7}{2}}{2 \cdot \frac{5}{2} \cdot \frac{7}{2}}$$

$$\frac{35}{4} = 5(5+1) - 1(1+1) + \frac{105}{4}$$

$$1(1+1) - 5(5+1) = \frac{35}{2} = \frac{70}{4} = \frac{72}{4} - \frac{2}{4}$$

Q3:

a.)

$$i\hbar \frac{\partial \Psi}{\partial t} = (\hat{H} + \hat{V}(t)) \Psi$$

$$\Rightarrow i\hbar [\dot{c}_1 e^{-i\omega_1 t} |\psi_1\rangle + \dot{c}_2 e^{-i\omega_2 t} |\psi_2\rangle] = c_1 e^{-i\omega_1 t} \hat{V} |\psi_1\rangle + c_2 e^{-i\omega_2 t} \hat{V} |\psi_2\rangle$$

$$|1\rangle = |\psi_1\rangle, |2\rangle = |\psi_2\rangle \quad \omega_n = \frac{E_n}{\hbar} \quad V_{ij} = \langle i | \hat{V} | j \rangle$$

⊗ $\langle 2 | \dots$:

$$i\hbar \dot{c}_2 e^{-i\omega_2 t} = c_1 e^{-i\omega_1 t} V_{21} + c_2 e^{-i\omega_2 t} V_{22}$$

by parity argument $V_{22} = 0$

$$\dot{c}_2 = c_1 e^{i(\omega_2 - \omega_1)t} \langle 1 | e^{i\omega_1 t} \hat{V} e^{-i\omega_1 t} | 2 \rangle \frac{1}{i\hbar}$$

$$= -i\Omega c_1 e^{i\omega_{21}t} \cos \omega t \quad c_1(t)$$

$$\Omega = \frac{eE_0}{\hbar} \langle 1 | x | 2 \rangle \quad \omega_{21} = \omega_2 - \omega_1$$

$$b.) \dot{c}_2 \sim -i\Omega \cos \omega t e^{i\omega_{21}t}$$

$$\sim -i\Omega \frac{1}{2} (e^{i\omega t} + e^{-i\omega t}) e^{i\omega_{21}t}$$

$$\sim \frac{i\Omega}{2} e^{i(\omega_{21} - \omega)t} \quad \text{as } |\omega_{21} - \omega| \ll \omega + \omega_{21}$$

by RWA drop $e^{i(\omega + \omega_{21})t}$ term

$$G = \int_0^T dt \frac{i\Omega}{2} e^{i\delta t}$$

$$\delta = \omega_{21} - \bar{\omega}$$

$$= \frac{\Omega}{2\delta} \left[e^{i\delta t} \right]_0^T$$

$$= \frac{\Omega}{2\delta} (e^{i\delta T} - 1)$$

$$= \frac{\Omega}{2\delta} e^{\frac{i\delta T}{2}} \left(e^{\frac{i\delta T}{2}} - e^{-\frac{i\delta T}{2}} \right)$$

$$= \frac{\Omega}{\delta} e^{\frac{i\delta T}{2}} \sin\left(\frac{\delta T}{2}\right)$$

$$|K_2|^2 = \frac{\Omega^2}{\delta^2} \sin^2\left(\frac{\delta T}{2}\right)$$

c.)

$G_2(t)$ similar in (b) $\delta = -\delta$

$$G_2(t) = \frac{\Omega_q}{\delta} \sin e^{\frac{i\delta t}{2}} \sin\left(\frac{\delta t}{2}\right) = \frac{\Omega}{2\delta} (e^{i\delta t} - 1)$$

$$G_3(t) \sim \frac{i\Omega_b \Omega_q}{2\delta} \left(e^{i\omega_b t} + e^{-i\omega_b t} \right)$$

$$\sim e^{-i\omega_{23}t} \left(e^{i\omega_{21}t} - 1 \right)$$

for all the phase only remain

$$e^{i\delta t}, e^{-i\delta t}$$

as other are faster rotating

$$\dot{c}_3(t) \sim \frac{i\Omega_b\Omega_a}{2\Delta} e^{-i(\omega_b + \omega_a - \omega_{z1} - \omega_{z3})t}$$

2/

$$c_3(t) \sim \frac{i\Omega_b\Omega_a}{2\Delta} e^{i(\omega_b - \omega_a + \omega_{z1} - \omega_{z3})t}$$

$$\sim \frac{i\Omega_b\Omega_a}{2\delta} e^{-i\delta't} \quad \underline{\delta' = (\omega_a - \omega_b) - \omega_{z1}}$$

$$c_3(t) = \int_0^t dt' \frac{i\Omega_b\Omega_a}{2\delta} e^{-i\delta't'}$$

$$= \frac{-i\Omega_b\Omega_a}{2\delta\delta'} (e^{-i\delta't} - 1)$$

$$= \frac{\Omega_b\Omega_a}{\delta\delta'} e^{-i\frac{\delta't}{2}} \sin\left(\frac{\delta't}{2}\right)$$

$$|c_3|^2 = \frac{(\Omega_b\Omega_a)^2}{(\delta\delta')^2} \sin^2\left(\frac{\delta't}{2}\right)$$

$$\beta = \frac{\delta'}{2} \quad W = \frac{2\Omega_b\Omega_a}{\Delta}$$

the shape w.r.t time of $|C_3|^2$ is still only a $\sin^2$ of detuning δ' but the maximum prob depends on both rabi oscillation rate and both detuning freq

d.)

for C_2 ~~at~~, by RWA only effective term should have $e^{\pm i \delta t}$ other term are oscillating too fast compare to this which average out
 $\times$ any combination of $\pm \omega_b + \omega_{z1}$ ~~is~~ can be Δ so legit to ignore ~~the~~ $\cos(\omega_b t)$ term

for C_3 term only with $e^{\pm i \delta t}$ should be effective. since C_2 only contain ω_a a ω_b need for the phase to be $\omega_a - (\omega_a - \omega_b) - \omega_{z1}$ so $\cos(\omega_a t)$ can be ignored, as all the term produce by this are oscillating too fast